

Subject: Indian Geography & Disaster Management / NCERT Class 11 Ch. 6 / UPSC Core

30-Second Snapshot

- A **natural hazard** is an environmental circumstance with the potential for harm, whereas a **natural disaster** causes sudden, widespread destruction beyond human coping capacity.
- Disasters are exacerbated by human activities such as deforestation, unscientific land use, and urban expansion in fragile zones.
- India is divided into **five seismic damage risk zones**, with the highest tectonic risk concentrated along the Himalayan arc and Northeast states.
- Tsunamis** act as shallow-water waves, amplifying in height up to 15 meters as they crash into shallow coastal regions.
- India experiences frequent **tropical cyclones** over the Bay of Bengal and Arabian Sea, driven by latent heat release and regulated by the Coriolis force.
- Approximately **19 percent of India's area** is drought-prone, categorized across meteorological, agricultural, hydrological, and ecological dimensions.
- Institutional response is anchored by the **Disaster Management Act (2005)** across pre, during, and post-disaster phases.

Core Concepts & Disaster Typologies

Disaster Category / Hazard	Operational Definition & Mechanics	Key Regional & Structural Signatures	Mitigation & Management Focus
Earthquakes	Sudden tectonic energy release along crustal faults.	Five damage risk zones; highest in Himalayas and Northeast states.	Earthquake-resistant building codes and micro-zoning mapping.
Tsunamis	Seismic sea waves generated by sea-floor displacement.	Long wavelength in deep ocean; towering shoaling waves in coastal zones.	International warning systems and coastal evacuation shelters.
Tropical Cyclones	Low-pressure atmospheric heat engines between 30°N and 30°S.	Form over Bay of Bengal and Arabian Sea; generate destructive storm surges.	Cyclone tracking radars, warning systems, and coastal afforestation.
Drought Typologies	Water deficit conditions spanning four distinct domains.	Spans meteorological, agricultural, hydrological, and ecological types.	Watershed management, rainwater harvesting, and drought-resistant crops.

Comparative Matrix: Vulnerability & Severity in India

- Seismic Vulnerability:**
 - High-Risk Zones:** Jammu & Kashmir, Ladakh, Himachal Pradesh, Uttarakhand, Sikkim, Northeast states, and Kuchchh (Gujarat). Stable Deccan Plateau is relatively safe.
- Flood & Drought Extents:**
 - Floods:** Covers roughly 40 million hectares, heavily affecting Assam, Bihar, and West Bengal.
 - Droughts:** Affects roughly 19 percent of India's area and 12 percent of population annually, with extreme aridity in western Rajasthan.
- Landslide Risk Zones:**
 - Very High:** Unstable young Himalayas, Andaman & Nicobar, and steep slopes of the Western Ghats.

Common UPSC Exam Traps

- Trap 1 (Hazard vs. Disaster):** A natural hazard is merely an environmental condition with the *potential* to cause harm (e.g., steep Himalayan slopes), whereas a disaster is the actual *realization* of widespread destruction and social disruption.
- Trap 2 (Purely Natural Disasters):** Disasters are rarely purely natural; human activities such as floodplain colonization, unscientific road cuts, and greenhouse gas emissions

actively intensify their frequency and impact.

- **Trap 3 (Tsunami Wave Speed and Height):** In the open ocean, tsunamis have extremely long wavelengths and low heights (barely noticeable), but they *shorten in wavelength and surge drastically in height* upon entering shallow coastal waters.
- **Trap 4 (Equatorial Cyclone Formation):** Tropical cyclones *cannot form within 5 degrees of the equator* because the Coriolis force is nearly zero, preventing the maintenance of low-pressure cores.
- **Trap 5 (Peninsular Earthquake Stability):** While the Peninsular block is generally stable, major earthquakes have occurred in stable shield regions (e.g., Latur and Koyna), proving that intra-plate fault systems can generate severe seismic events.

High-Yield Facts Box

Disaster Feature	Governing Mechanism	Diagnostic Identifier
Yokohama Strategy	International framework (1994)	Formalized global commitment to disaster risk reduction during the IDNDR decade.
Bay of Fundy	Tidal resonance	Location of the world's highest tidal range (15 to 16 meters).
Majuli (Assam)	Riverine deposition	World's largest riverine island, benefiting from annual Brahmaputra silt deposition despite flood losses.
Trikal Conditions	Combined drought impact	Simultaneous scarcity of food (akal), fodder (trinkal), and drinking water (jalkal).
Disaster Management Act	National legislation (2005)	Establishes legal frameworks and the National Institute of Disaster Management in India.

Prelims Practice Challenge

Q1. Consider the following statements regarding natural hazards and disasters in India:

1. A natural hazard is an environmental circumstance with the potential to cause harm, whereas a natural disaster results in large-scale widespread destruction beyond human coping capacity.
2. Tropical cyclones can readily form within 2 degrees of the equator because the Coriolis force is at its maximum strength in equatorial regions.
3. The Deccan Plateau is generally classified as a relatively safe, stable zone in India's earthquake damage risk mapping.
4. Tsunamis travel as shallow-water waves whose height increases dramatically as they approach shallow coastal areas.

Which of the statements given above are correct?

- (A) 1, 3, and 4 only
- (B) 2 and 3 only
- (C) 1, 2, and 4 only
- (D) 1, 2, 3, and 4

Answer: (A) 1, 3, and 4 only

Explanation: Statement 1 is correct because hazards represent potential harm while disasters represent actual catastrophic events. Statement 2 is incorrect because tropical cyclones *cannot form within 5 degrees of the equator* due to negligible Coriolis force. Statements 3 and 4 are correct descriptions of Deccan stability and coastal tsunami shoaling.

Q2. With reference to drought classifications and disaster management in India, consider the following statements:

1. Meteorological drought is defined strictly by prolonged inadequate rainfall poorly distributed over time and space.
2. The Yokohama Strategy and Plan of Action for a Safer World was formalized at a UN conference in May 1994.
3. The Disaster Management Act was enacted in India in the year 2005 to establish structured institutional response mechanisms.
4. Agricultural drought occurs when water availability in deep subterranean aquifers falls below

historical precipitation recharge levels.

Which of the statements given above are correct?

- (A) 1 and 4 only
- (B) 1, 2, and 3 only
- (C) 2, 3, and 4 only
- (D) 1, 2, 3, and 4

Answer: (B) 1, 2, and 3 only

Explanation: Statements 1, 2, and 3 are correct definitions of meteorological drought, the 1994 Yokohama Strategy, and the 2005 Disaster Management Act. Statement 4 is incorrect because *hydrological drought* refers to depleted water levels in aquifers, lakes, and reservoirs, whereas *agricultural drought* refers specifically to low soil moisture causing crop failure.

Mains Practice Question (10 Marks / 150 Words)

Question: "Distinguish between natural hazards and disasters, highlighting how anthropogenic factors intensify disaster vulnerability in India. Examine the institutional framework established under the Disaster Management Act of 2005."

Structured Answer Framework:

- **Introduction (25–30 words):**

Define a natural hazard as an environmental potential for harm and a natural disaster as a catastrophic disruption of social systems, noting India's high multi-hazard vulnerability profile.

- **Body Arguments (80–90 words):**

- *Anthropogenic Intensification:* Explain how human activities—such as colonizing floodplains, unscientific construction in fragile Himalayan slopes, coastal port development, and deforestation—turn natural hazards into severe human-made or intensified disasters.
- *Institutional Framework (DM Act 2005):* Discuss the provisions of the Disaster Management Act of 2005, which established statutory bodies at national, state, and district levels, created the National Institute of Disaster Management, and shifted the policy paradigm from post-disaster relief to pre-disaster preparedness and mitigation.

- **Conclusion / Way Forward (25–30 words):**

Effective disaster management requires integrating scientific vulnerability mapping, strict land-use zoning, and community-based preparedness to build long-term national resilience.